MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 13: Complex Series - examples

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Lecture outline

1. Examples

Example 1

Example

Prove that if $\sum \alpha_n$ converges then it is Cauchy convergent.

Proof: Suppose $\sum \alpha_n$ converges. Then $\forall \varepsilon > 0$, $\exists N(\varepsilon) \in Z^+$ s.t. $n, m > N(\varepsilon) \Rightarrow |A - A_n| < \frac{\varepsilon}{2}$ and $|A - A_m| < \frac{\varepsilon}{2}$. But

$$|A_n - A_m| = |A_n - A + A - A_m| \leq |A_n - A| + |A - A_m| = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

■

Example 2. Part 1

Consider the series

$$\sum_{n=1}^{\infty} z^n (1-z)$$

Show that the series converges for $|z| < 1$ and find its sum.

Proof: Given $\epsilon < 1$, assume $z \neq 0$.

$$\begin{aligned} S_n &= z(1-z) + z^2(1-z) + \cdots + z^n(1-z) \\ &= z - z^{n+1} \end{aligned}$$

Thus

$$|S_n - z| = \left|-z^{n+1}\right| = |z|^{n+1} < \epsilon \tag{1}$$

for $(n+1)\ln|z| < \ln\epsilon$.

Example 2. Part 2

This means that we require

$$n + 1 > \frac{\ln \epsilon}{\ln |z|}.$$

Now since $|z| < 1$, $\ln |z| < 0$ and $\frac{\ln \epsilon}{\ln} > 0$.
Thus (1) will hold if we take

$$n > \frac{\ln \epsilon}{\ln |z|} - 1$$

For $z = 0$, $S_n = 0 \implies |S_n - 0| < \epsilon$ for all n.
■

Example 3. Part 1

Consider the series (same as in Example 2)

$$\sum_{n=1}^{\infty} z^n(1-z)$$

Prove that the series converges uniformly to its sum for $|z| \leq \frac{1}{2}$.
(Hint: Get a lower bound on N.)

Proof: We have that

$$|S_n - z| < \epsilon \quad \text{for } n > \frac{\ln \epsilon}{\ln |z|} - 1$$

Thus the series converges to z for $|z| < 1$ and certainly for $|z| \leq \frac{1}{2}$. If $|z| \leq \frac{1}{2}$,

Example 3. Part 2

then the largest value of

$$\frac{\ln \epsilon}{\ln |z|} - 1$$

occurs when $|z| = \frac{1}{2}$ and is given by

$$\frac{\ln \epsilon}{\ln \frac{1}{2}} - 1 = -\frac{\ln \epsilon}{\ln 2} - 1$$

Take $N = N(\epsilon)$ to be an integer greater than $-\frac{\ln \epsilon}{\ln 2} - 1$. Then $|S_n - z| < \epsilon$ for $n > N$ and $|z| \leq \frac{1}{2}$. ■

Example 4.

Consider the series (same as in Example 2)

$$\sum_{n=1}^{\infty} z^n(1-z)$$

Does the series converge uniformly for $|z| \leq 1$? Explain.

The argument presented in Example 3 applies if $z = .9$ or $.99$. We can use any integer greater than $\frac{\ln \epsilon}{\ln .9} - 1$ or any integer greater than $\frac{\ln \epsilon}{\ln .99} - 1$. However, as we get closer to 1 we see that we are approaching $\frac{\ln \epsilon}{\ln 1} - 1$, which is infinite. Thus no finite value of N exists for $z = 1$. Therefore the series does not converge uniformly for $|z| \leq 1$.■

Example 5.

Consider the series (same as in Example 2)

$$\sum_{n=1}^{\infty} z^n(1-z)$$

Prove that the series is absolutely convergent for $|z| < 1$.

Proof: Using the ratio test, $\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{z^{n+1}(1-z)}{z^n(1-z)}\right| = |z| \to |z|$ as $n \to \infty$.
Thus $\sum_{n=1}^{\infty} z^n(1-z)$ is absolutely convergent for $|z| < 1$. ■

Example 6.

Let

$$f(z) = \sum_{n=1}^{\infty} z^n (1 - z).$$

Determine $f(z)$ and show that it is discontinuous at $z = 1$.

$S_n = z - z^{n+1}$. If $|z| < 1$, then $\lim_{n\to\infty} S_n = z$. If $z = 1$, then $S_n = 0$.

$$f(z) = \begin{cases} z & \text{if} \quad |z| < 1 \\ 0 & \text{if} \quad z = 1 \\ \text{Does not exist} & \quad \text{otherwise} \end{cases}$$

Thus for $z = 1$, $\lim_{n\to\infty} S_n = 0$. Thus $f(z)$ is discontinuous at $z = 1$.■

Example 7.

Example

Show that $S_n = 1 + z + z^2 + \cdots + z^{n-1} = \dfrac{1 - z^n}{1 - z}$.

Proof: Consider $z \neq 1$.
$S_n = 1 + z + z^2 + \cdots + z^{n-1} \Rightarrow z \cdot S_n = z + z^2 + \cdots + z^{n-1} + z^n$
$S_n - z \cdot S_n = S_n(1 - z) = 1 - z^n$. Dividing by $1 - z$, we get $S_n = \dfrac{1 - z^n}{1 - z}$. ■

Example 8.

Example

Find a series expansion for $\dfrac{1}{1+z}$ about 0.

$$\frac{1}{1+z} = \frac{1}{1-(-z)} = \sum_{n=0}^{\infty}(-1)^n z^n$$

if $|z| < 1$

Example 9.

Example

Expand $\dfrac{1}{1+z^3}$ about $z=0$. Give the radius of convergence.

Let $t = z^3$. Then

$$\frac{1}{1+z^3} = \frac{1}{1+t} = \sum_{n=0}^{\infty}(-1)^n t^n = \sum_{n=0}^{\infty}(-1)^n z^{3n},$$

if $|z| < 1$

Example 10.

Example

Find the first three nonzero terms in the Taylor series expansion of $\tan z$ about $z = 0$.

Let $f(z) = \tan z$
Then $f'(z) = \sec^2 z$,
$f''(z) = 2(\tan z)\left(1 + \tan^2 z\right)$,
$f'''(z) = 2 + 8\tan^2 z + 6\tan^4 z$,
$f^{(4)}(z) = 16\tan z + 40\tan^3 z + 24\tan^5 z$,
and $f^{(5)}(z) = 136\tan^2 z + 240\tan^4 z + 120\tan^6 z + 16$
So

$$\tan z = 0 + z + 0 + \frac{2z^3}{3!} + 0 + \frac{16z^5}{5!} = z + \frac{z^3}{3} + \frac{2z^5}{15} + \cdots$$

Example 11.

Example

Expand $\dfrac{1}{1+z}$ about $z=-2i$. Give the region of convergence.

$$\frac{1}{1+z}=\frac{1}{1-2i+(z+2i)}=\frac{1}{1-2i}\frac{1}{\left(1+\dfrac{z+2i}{1-2i}\right)}=\frac{1+2i}{5}\frac{1}{\left(1+\dfrac{z+2i}{1-2i}\right)}$$

Let $t=\dfrac{z+2i}{1-2i}$. Then

$$\frac{1+2i}{5}\frac{1}{\left(1+\dfrac{z+2i}{1-2i}\right)}=\frac{1+2i}{5}\frac{1}{1+t}=\frac{1+2i}{5}\sum_{n=0}^{\infty}(-1)^n t^n, \text{ if } |t|<1$$

So we have

$$\frac{1+2i}{5}\sum_{n=0}^{\infty}(-1)^n\left(\frac{z+2i}{1-2i}\right)^n=\frac{1+2i}{5}\sum_{n=0}^{\infty}(-1)^n\left(\frac{1+2i}{5}\right)^n(z+2i)^n=$$

$$\sum_{n=0}^{\infty}(-1)^n\left(\frac{1+2i}{5}\right)^{n+1}(z+2i)^n.$$

This series converges for $\left|\dfrac{z+2i}{1-2i}\right|<1 \Leftrightarrow |z+2i|<\sqrt{5}$.

Example 12.

Example

Let $f(z) = \log(1+z)$, where $f(0) = 0$.

(a) Expand $f(z)$ in a Taylor series about $z = 0$.

$f'(z) = \dfrac{1}{1+z} = \sum_{n=0}^{\infty}(-1)^n z^n$ for $|z| < 1$

$$log(z+1) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{n+1}}{n+1}$$

(b) Determine the region of convergence of the series found in (a).

The antiderivative, $f(z)$ will have the same radius of convergence as $f'(z)$. Thus series $f(z)$ converges if $|z| < 1$

Example 13

Example

Expand $\log \dfrac{1+z}{1-z}$ in a Taylor series about $z = 0$.

Let $f(z) = \log \dfrac{1+z}{1-z}$.
Then

$$f'(z) = -\frac{2}{(1+z)(-1+z)} = \frac{1}{1+z} - \frac{1}{-1+z} = \frac{1}{1+z} + \frac{1}{1-z} =$$

$$\sum_{n=0}^{\infty}(-1)^n z^n + \sum_{n=0}^{\infty} z^n = \sum_{n=0}^{\infty} 2z^{2n}$$

So $f(z) = \sum_{n=0}^{\infty} \dfrac{2z^{2n+1}}{2n+1}$

Thank you